

Welcome to STA2002!

Fall Term 2023-24

Today/this week's class

- ❑Course organization
- ❑Introduction: Probability versus Statistics
- ❑Review of STA2001
  - ❑Common distributions
  - ❑Distributions related to the normal distribution
  - ❑Limit theorems

Course staff

- **Instructor:**
  - *Fangda Song (Weeks 1-3)*
    - **Office:** Rm 519, Daoyuan Bldg
    - **Email:** songfangda@cuhk.edu.cn
  - *Yongtao Guan (Weeks 4-14)*
    - **Office:** Rm 509a, Daoyuan Bldg
    - **Email:** guanyongtao@cuhk.edu.cn
- Office hour: Mon 10:30 am – 11:30 am

Course staff

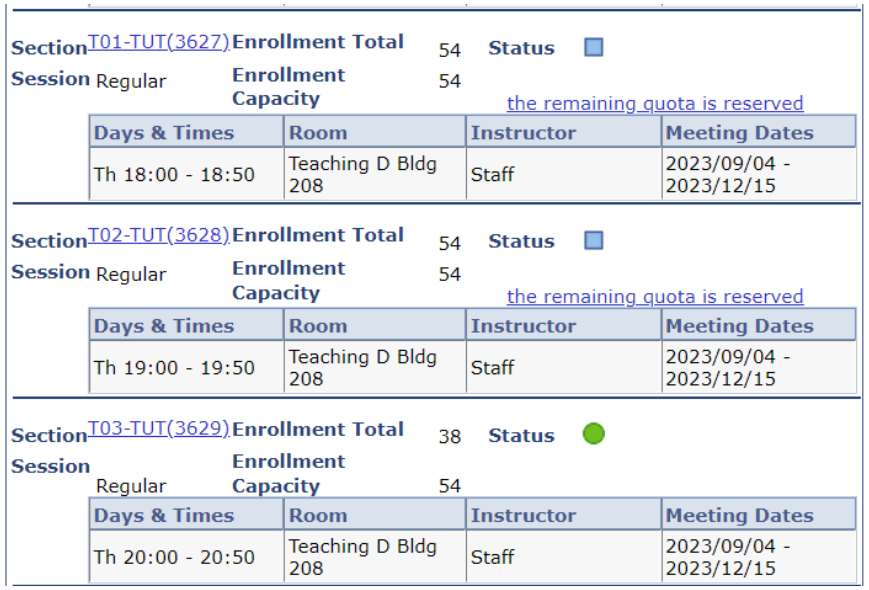
- **Session 1**:
  - Room: Rm 101, Chengdao Bldg
  - Time: Mon Wed 8:30 am– 9:50 am
- **Session 2**:
  - Room: Rm 102, Chengdao Bldg
  - Time: Mon Wed 3:30 pm – 4:50 pm

Course staff

- **TA:**
  - Kang Gong
    - **Office:** Rm 503, Daoyuan Bldg
    - **Email:** kanggong@link.cuhk.edu.cn
  - *Qian, Zishan*
    - **Office:** #11, The smaller room of SDS Research Lab (4th Floor), Zhi Xin Bldg
    - **Email:** 222042009@link.cuhk.edu.cn
  - *Yang, Yiran*
    - **Office:** #21, The smaller room of SDS Research Lab (4th Floor), Zhi Xin Bldg
    - **Email:** 119010386@link.cuhk.edu.cn
  - *Liu, Shengjie*
    - **Office:** #54, The larger room of SDS Research Lab (4th Floor), Zhi Xin Bldg
    - **Email:** 221049041@link.cuhk.edu.cn

Tutorials

- Tutorials begin on the **second** week, i.e. the week of Sept 11st



Section T01-TUT(3627)	Enrollment Total	54	Status
Session Regular	Enrollment Capacity	54	
	the remaining quota is reserved		

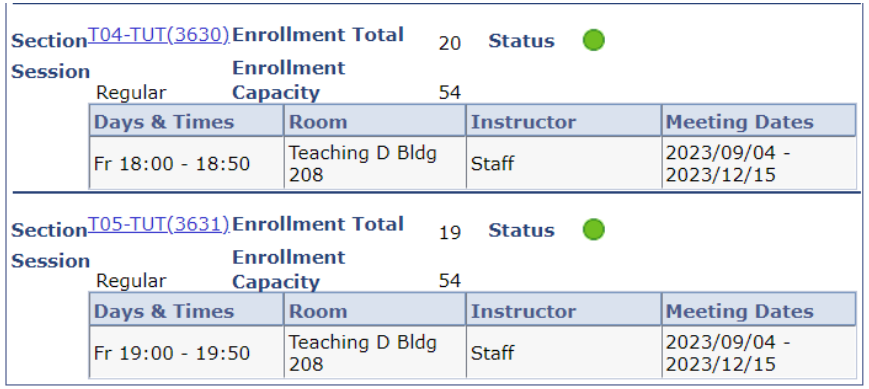
Days & Times	Room	Instructor	Meeting Dates
Th 18:00 - 18:50	Teaching D Bldg 208	Staff	2023/09/04 - 2023/12/15

Section T02-TUT(3628)	Enrollment Total	54	Status
Session Regular	Enrollment Capacity	54	
	the remaining quota is reserved		

Days & Times	Room	Instructor	Meeting Dates
Th 19:00 - 19:50	Teaching D Bldg 208	Staff	2023/09/04 - 2023/12/15

Section T03-TUT(3629)	Enrollment Total	38	Status
Session Regular	Enrollment Capacity	54	

Days & Times	Room	Instructor	Meeting Dates
Th 20:00 - 20:50	Teaching D Bldg 208	Staff	2023/09/04 - 2023/12/15



Section T04-TUT(3630)	Enrollment Total	20	Status
Session Regular	Enrollment Capacity	54	

Days & Times	Room	Instructor	Meeting Dates
Fr 18:00 - 18:50	Teaching D Bldg 208	Staff	2023/09/04 - 2023/12/15

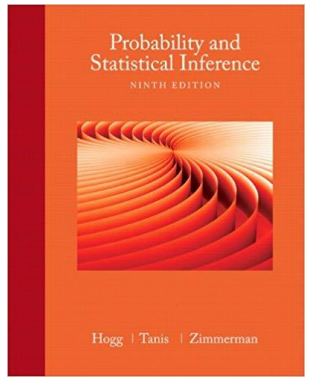
Section T05-TUT(3631)	Enrollment Total	19	Status
Session Regular	Enrollment Capacity	54	

Days & Times	Room	Instructor	Meeting Dates
Fr 19:00 - 19:50	Teaching D Bldg 208	Staff	2023/09/04 - 2023/12/15

Course contents

- Applications of laws of large numbers and central limit theorem in parameter estimation and test statistic construction.
- Point estimation, maximum likelihood estimation, confidence intervals.
- Hypothesis testing: basic concepts and more advanced techniques.
- Regression analysis, contingency tables, analysis of variance.
- Bayesian estimation (if time permits)

Course resources



- **Text:** Hogg, R. V., Tanis, E. A. Zimmerman, D. L. (2015) Probability and Statistical Inference, 9th edition, Pearson

- We will be cover most of the material from Chapter 6 to Chapter 9.

- In our first week of class, we will quickly review Chapter 1 to Chapter 5, covered in STA2001. For those of you who did not take STA2001 and still would like to take this class, please make sure you understand these material.

- Blackboard site: https://bb.cuhk.edu.cn

Assessment Scheme

Component/ method	Weight
Assignments	25
Mid-Term Exam	30
Final Exam	45

- **Course Policies:**

✓ There are 6 assignments. You are expected to work on all assignments on your own. **No late assignment will be accepted.** No exceptions. **The lowest assignment score will be dropped.**

✓ Midterm exam allows one cheat sheet, while final exam does up to two cheat sheets. The cheat sheets should **be handwritten and in A4 size**.

✓ A makeup exam is allowed ONLY under provable unforeseen circumstances beyond the control of the student. Only a written request for a makeup exam that is sent BEFORE the exam date and is accompanied by all relevant supporting documentation will be considered.

✓ Any form of academic dishonesty, which includes, but is not limited to, cheating in examinations, plagiarism and sharing of assignments with others, **will NOT be tolerated**. Any related offence will lead to disciplinary action including termination of studies at the University.

Important notes

Throughout this course, you are not only learning to correctly apply statistical techniques, you are also learning how to correctly formulate proofs and derivations in statistics.

A correct use of notation (that has to be introduced before it can be used!) and a correct derivation is at least as important as the correct answer!

Tentative outline

Week	Content/ topic/ activity
1	Review of families of distributions: Gamma, Normal, Exponential, Chi-square, t-distribution, F-distribution, Bernoulli, Binomial, etc. Moment generating function; Central Limit Theorem.
2	Maximum Likelihood Estimation; Method of moments; Unbiased estimation.
3	Confidence interval (CI) for means; CI for the difference of two means; CI for proportions.
4	Tests of statistical hypotheses, Critical region, Significance level, p-value, Power, t-test, F-test.
5	More t-test; Tests about proportions.
6	Sample size; Power of a statistical test; Order statistics; QQ plots.
7	Review Session. **Midterm Exam**.
8	Distribution-free CIs for Percentiles; The Wilcoxon tests.
9	Chi-square goodness-of-fit tests; Contingency tables.
10	Testing for homogeneity; Testing for independence; One-way ANOVA.
11	Two-way ANOVA; Introduction to regression.
12	More regression; Tests concerning regression and correlation.
13	Likelihood ratio tests.
14	Review Session. **Final Exam**.

Today/this week's class

☑Course organization

❑Introduction: Probability versus Statistics

❑Review of STA2001

- ❑Common distributions
- ❑Distributions related to the normal distribution
- ❑Limit theorems

Probability versus Statistics

Probability

- Given a *model*, figure *out* *what the model says should happen*

Statistics

- Given *observations* of what happened, figure out

  what model explains the observations

  How much uncertainty we have when using this model for prediction

Probability versus Statistics: examples

Flip a coin five times.

- Probability:
  - Assuming the coin is a fair coin, what is the probability of observing the same side coming up all five times?

- Statistics:
  - You flip the coin five times and see it landed on Heads each time. Should you conclude the coin is not a fair coin?

  - You flip the coin 20 times and see Heads 14 times. What is your estimate of the probability of landing on Heads?
  - What if you flip the coin 1000 times and see Heads 700 times? Do you feel equally confident about the estimates in these two scenarios?

Probability versus Statistics: examples

A candy maker produces mints with label weight 20.4 grams.

- Probability:
  - Assuming the weights are normally distributed, with mean equal to the label weight and variance 0.16 $gram^2$, what is the probability that a randomly selected mint weighs at most 19.9 gram?

- Statistics:
  - A random sample of 25 mints gives an average weight of 19.9 grams. What can we conclude about the weight of the mints?

Probability versus Statistics

The previous examples illustrated the following key idea:

In statistics, we *compare statistical measurements with what probability predicts*, and use that to make inferences about the probabilistic model (along with information on how much confidence we have in these inferences)

Probability versus Statistics

Flip a coin 20 times and find 14 heads ➔

The coin is a fair coin? Unlikely

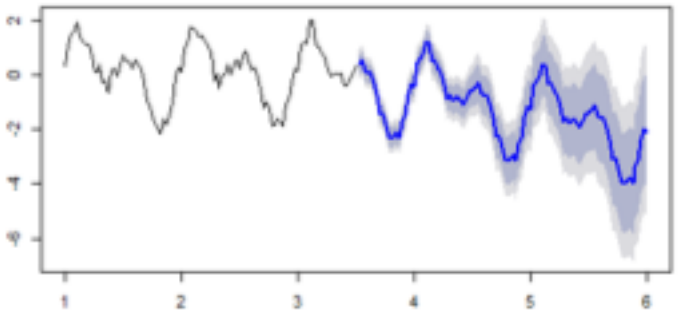
The coin has a P(head)=0.6? Possibly

The coin has a P(head)=0.7? More likely than above

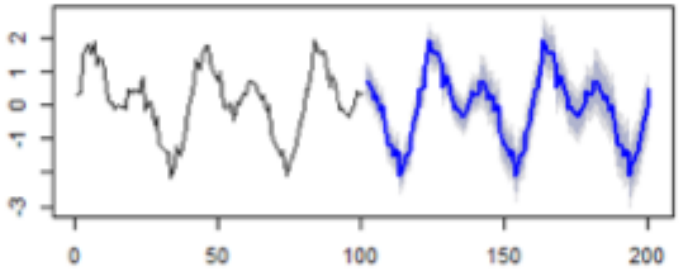
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How do we quantify our belief about the coin, after seeing the results of the coin tosses?

The place of statistics in data science



Statistics provides:



Unique way of thinking

Tools that quantify uncertainty

Quantify loss/gain from decisions while considering uncertainty

Its thinking penetrates all disciplines that consider modeling and prediction

The place of statistics in data science



Where does uncertainty come from:



Sometimes it is the lack of information (data)

Sometimes complete data collection is infeasible

Example: Predict a person's height – need to know all genetics and life exposure (food, exercise, weather .....)

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☑Introduction: Probability versus Statistics

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- ❑Limit theorems

Review of STA2001

Common Distributions

Random variable

A **random variable** is a function that assigns numeric values to different events in a sample space.

Two types of random variables are discussed in this text: discrete and continuous.

A random variable for which there exists a discrete set of numeric values is a **discrete random variable.**

A random variable whose possible values cannot be enumerated is a **continuous random variable.**

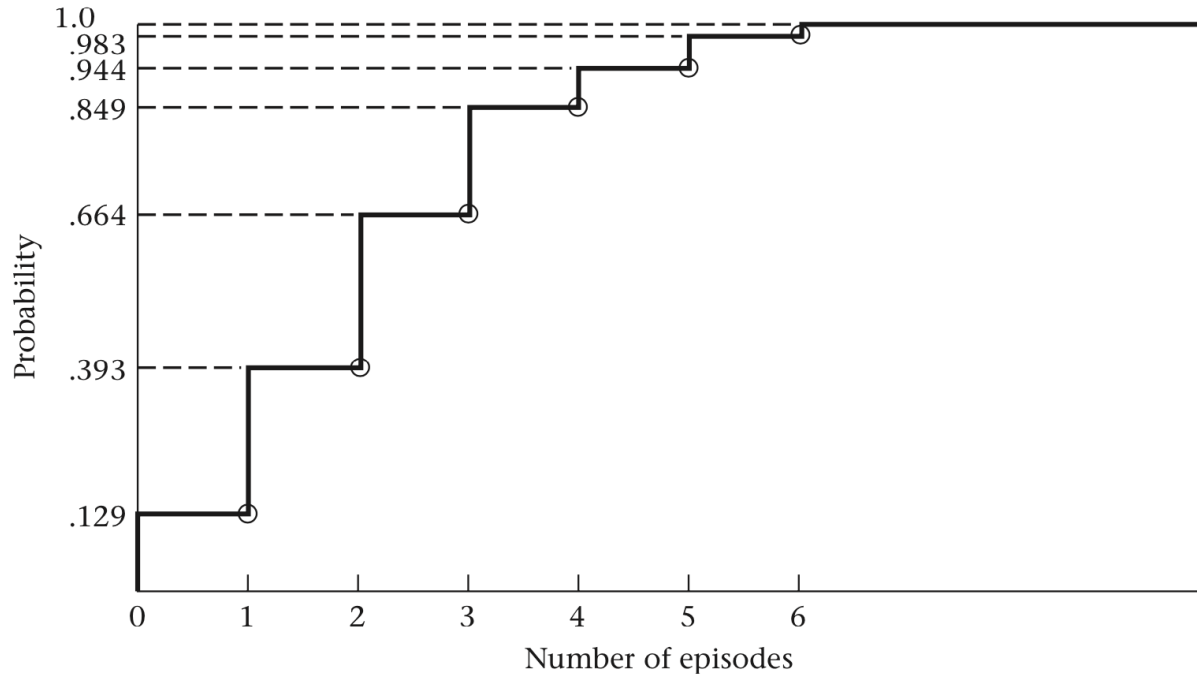
Discrete Random variable

Probability-mass function for the number of episodes of otitis media in the first 2 years of life

r	0	1	2	3	4	5	6
$Pr(X = r)$	.129	.264	.271	.185	.095	.039	.017

The **cumulative-distribution function (cdf)** of a random variable X is denoted by $F(X)$ and, for a specific value x of X, is defined by $Pr(X \leq x)$ and denoted by $F(x)$.

Cumulative-distribution function for the number of episodes of otitis media in the first 2 years of life



$$
\begin{aligned}
F(x) &= 0 && \text{if} \quad x < 0 \\
F(x) &= .129 && \text{if} \quad 0 \leq x < 1 \\
F(x) &= .393 && \text{if} \quad 1 \leq x < 2 \\
F(x) &= .664 && \text{if} \quad 2 \leq x < 3 \\
F(x) &= .849 && \text{if} \quad 3 \leq x < 4 \\
F(x) &= .944 && \text{if} \quad 4 \leq x < 5 \\
F(x) &= .983 && \text{if} \quad 5 \leq x < 6 \\
F(x) &= 1.0 && \text{if} \quad x \geq 6
\end{aligned}
$$

Discrete Random variable

The **expected value of a discrete random variable** is defined as

$$E(X) \equiv \mu = \sum_{i=1}^{R} x_i Pr(X = x_i)$$

Probability-mass function for the number of episodes of otitis media in the first 2 years of life

r	0	1	2	3	4	5	6
$Pr(X = r)$	.129	.264	.271	.185	.095	.039	.017

$$E(X) = 0(.129) + 1(.264) + 2(.271) + 3(.185) + 4(.095) + 5(.039) + 6(.017) = 2.038$$

Discrete Random variable

The **variance of a discrete random variable**, denoted by $Var(X)$, is defined by

$$Var(X) = \sigma^2 = \sum_{i=1}^{R} (x_i - \mu)^2 Pr(X = x_i)$$

$$\sigma^2 = E(X-\mu)^2 = \sum_{i=1}^{R} x_i^2 Pr(X = x_i) - \mu^2$$

Probability-mass function for the number of episodes of otitis media in the first 2 years of life

r	0	1	2	3	4	5	6
$Pr(X = r)$	.129	.264	.271	.185	.095	.039	.017

$$\begin{aligned}\sum_{i=1}^{R} x_i^2 Pr(X = x_i) &= 0^2(.129) + 1^2(.264) + 2^2(.271) + 3^2(.185) \\ &\quad + 4^2(.095) + 5^2(.039) + 6^2(.017) \\ &= 0(.129) + 1(.264) + 4(.271) + 9(.185) \\ &\quad + 16(.095) + 25(.039) + 36(.017) \\ &= 6.12\end{aligned}$$

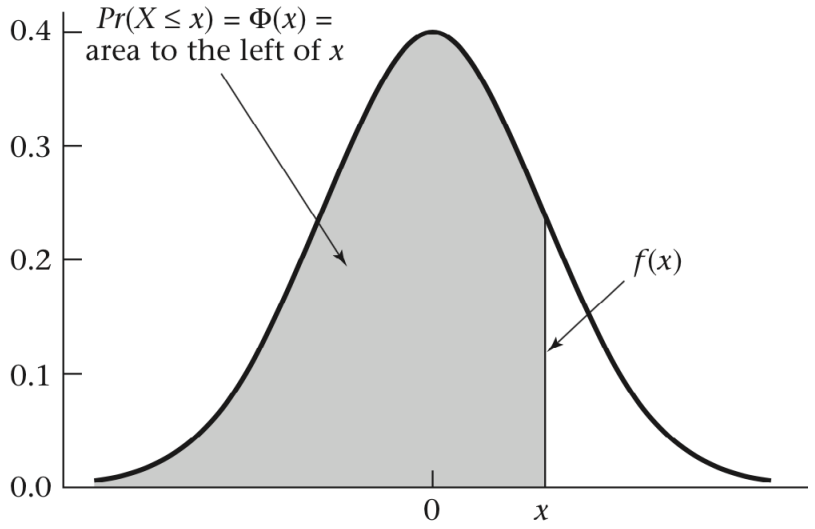
$$Var(X) = \sigma^2 = 6.12 - (2.038)^2 = 1.967$$

Continuous Random variable

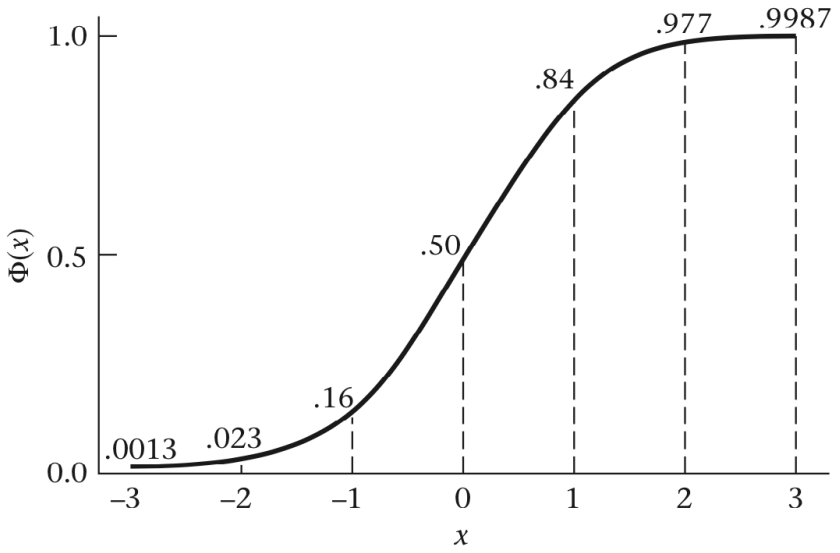
Definition 5.1 The **probability-density function** of the random variable X is a function such that the area under the density-function curve between any two points a and b is equal to the probability that the random variable X falls between a and b. Thus, the total area under the density-function curve over the entire range of possible values for the random variable is 1.

Definition 5.2 **The cumulative-distribution function** for the random variable X evaluated at the point a is defined as the probability that X will take on values $\leq a$. It is represented by the area under the pdf to the left of a.

The cdf [$\Phi(x)$] for a standard normal distribution



The cdf for a standard normal distribution [$\Phi(x)$]



Statistical independence

Independence is simple, dependence can be complicated

Variables *X* and *Y* , taking on possible values $x_1, x_2, \ldots,$ and $y_1, y_2, \ldots,$

In the discrete case,

independent if, for all i and j,

$$P(X = x_i \text{ and } Y = y_j) = P(X = x_i)P(Y = y_j)$$

In the continuous case,

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Mean and variance

If $X_1, \ldots, X_n$ are jointly distributed random variables with expectations $E(X_i)$ and Y is a linear function of the X_i, $Y = a + \sum_{i=1}^{n} b_i X_i$, then

$$E(Y) = a + \sum_{i=1}^{n} b_i E(X_i)$$

If $\mathrm{Var}(X)$ exists and $Y = a + bX$, then $\mathrm{Var}(Y) = b^2 \mathrm{Var}(X)$.

The variance of X, if it exists, may also be calculated as follows:

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Mean and variance

If X and Y are jointly distributed random variables with expectations μ_X and μ_Y, respectively, the covariance of X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

provided that the expectation exists. ■

$$\text{Var}(a + \sum_{i=1}^{n} b_i X_i) = \sum_{i=1}^{n} \sum_{j=1}^{n} b_i b_j \text{Cov}(X_i, X_j).$$ ■

$$\text{Var}(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} \text{Var}(X_i), \text{ if the } X_i \text{ are independent.}$$ ■

Distributions - binomial

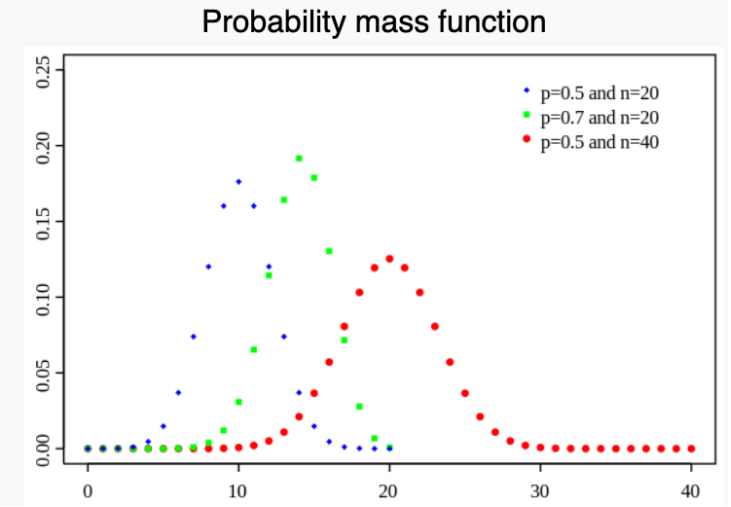


Consider flipping n coins, each of which has (independent) probability θ of coming up heads, and probability $1 - \theta$ of coming up tails($0 < \theta < 1$). Let X be the total number of heads.

$$p_X(x) = P(X = x) = \binom{n}{x} \theta^x (1-\theta)^{n-x} = \frac{n!}{x!\,(n-x)!} \theta^x (1-\theta)^{n-x}$$

$$x = 0, 1, 2, \ldots, n$$

$$E(X) = n\theta$$
$$var(X) = n\theta(1-\theta)$$



What is the probability of obtaining 2 boys out of 5 children if the probability of a boy is .51 at each birth and the sexes of successive children are considered independent random variables?

$$Pr(X = 2) = \binom{5}{2} (.51)^2 (.49)^3 = \frac{5 \times 4}{2 \times 1} (.51)^2 (.49)^3$$

$$= 10(.51)^2 (.49)^3 = .306$$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Distributions - Poisson

If we flip a *very large* number of coins (*n*), and each coin has a *very small* probability to come up heads $\theta = \lambda/n$

$$P(X = x) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \longrightarrow \lim_{n\to\infty} P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Poisson(λ) distribution

$$E(X) = \lambda$$
$$var(X) = \lambda$$

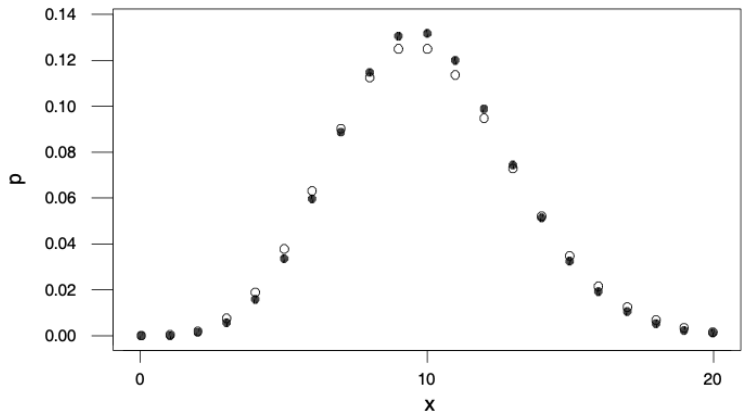
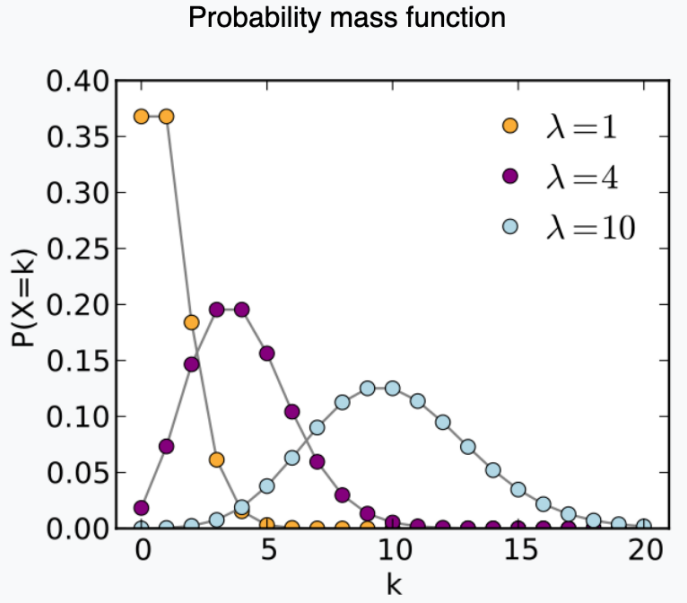


Figure 2.3.5: Plot of the Binomial$(100, 1/10)$ (● ● ●) and the Poisson(10) (○ ○ ○) probability functions at the values $0, 1, \ldots, 20$.



Distributions - geometric

Consider repeatedly flipping a coin that has probability θ of coming up heads, let *X* be the number of tails before the first head.

$$p_X(k) = (1-\theta)^k\theta, \;\; k = 0, 1, 2, 3, \ldots .$$

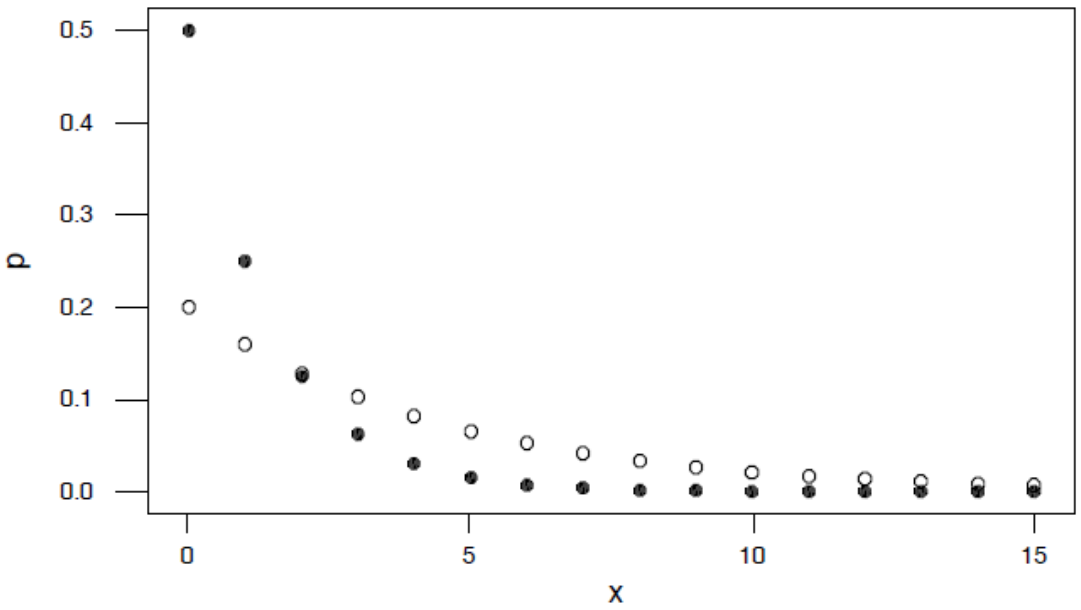


Figure 2.3.2: Plot of the Geometric$(1/2)$ (● ● ●) and the Geometric$(1/5)$ (○ ○ ○) probability functions at the values $0, 1, \ldots, 15$.

$$E(X) = \frac{1}{\theta}$$

$$var(X) = \frac{1-\theta}{\theta^2}$$

Distributions – negative binomial

Consider repeatedly flipping a coin that has probability θ of coming up heads, let *Y* be the number of tails that appear before the *r* th head.

$$p_Y(k) = \binom{r-1+k}{r-1}\theta^{r-1}(1-\theta)^k\theta = \binom{r-1+k}{k}\theta^r(1-\theta)^k$$

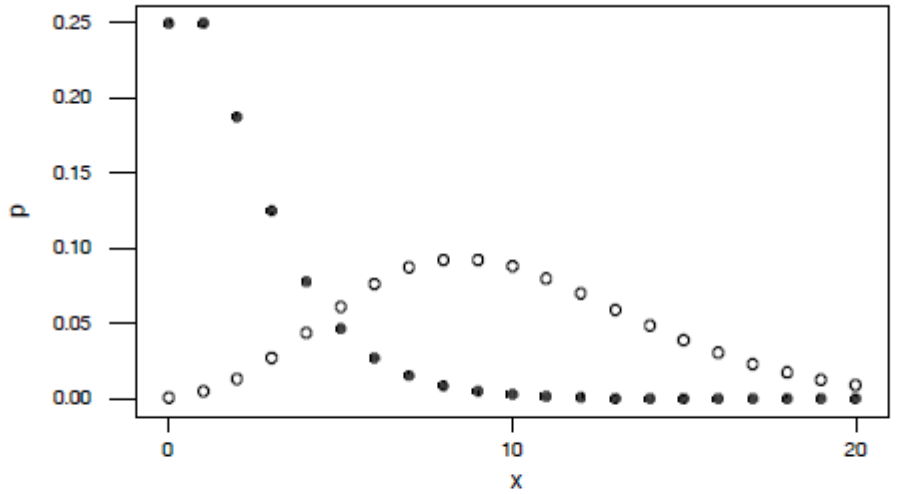


Figure 2.3.3: Plot of the Negative-Binomial$(2, 1/2)$ (● ● ●) probability function and the Negative-Binomial$(10, 1/2)$ (○ ○ ○) probability function at the values $0, 1, \ldots, 20$.

Distributions - hypergeometric

Suppose that an urn contains M white balls and $N - M$ black balls. Suppose we draw $n \leq N$ balls from the urn randomly. Each subset of n balls has the same probability of being drawn.

What is the probability that x white balls are obtained?

$$P(X = x) = \binom{M}{x}\binom{N-M}{n-x} \Big/ \binom{N}{n}$$

Hypergeometric(N, M, n)

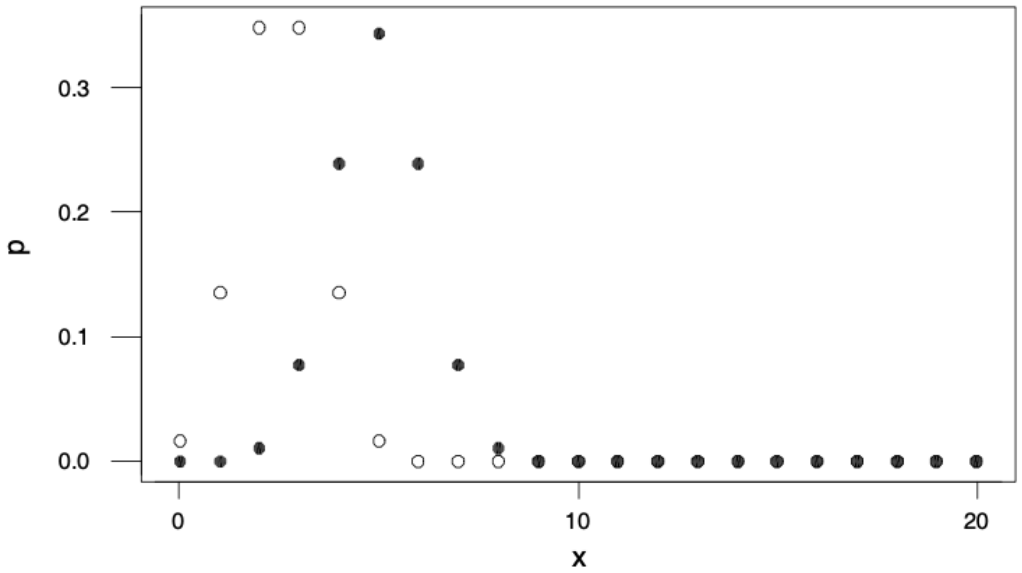


Figure 2.3.6: Plot of Hypergeometric(20, 10, 10) (● ● ●) and Hypergeometric(20, 10, 5) (○ ○ ○) probability functions.

Distributions - Uniform

Let L and R be any two real numbers with $L < R$,

whenever $L \le a \le b \le R$,

$$P(a \le X \le b) = \frac{b-a}{R-L}$$

$$P(X < L) = P(X > R) = 0.$$

$$f(x) = \begin{cases} \frac{1}{R-L} & L \le x \le R \\ 0 & \text{otherwise.} \end{cases}$$

$$E(X) = \frac{R+L}{2}$$

$$var(X) = \frac{1}{12}(R-L)^2$$

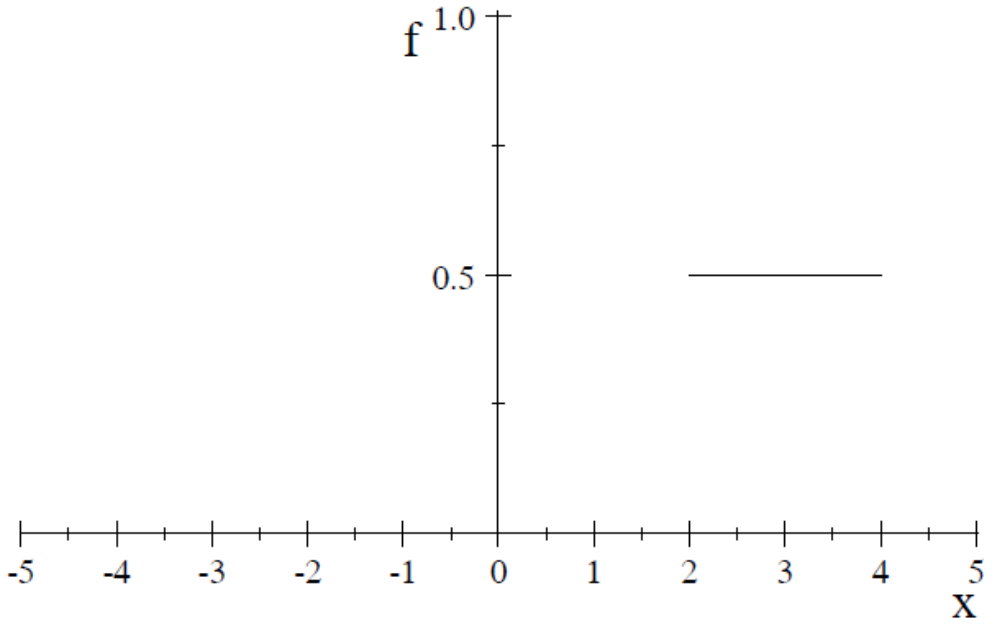


Figure 2.4.1: A Uniform[2, 4] density function.

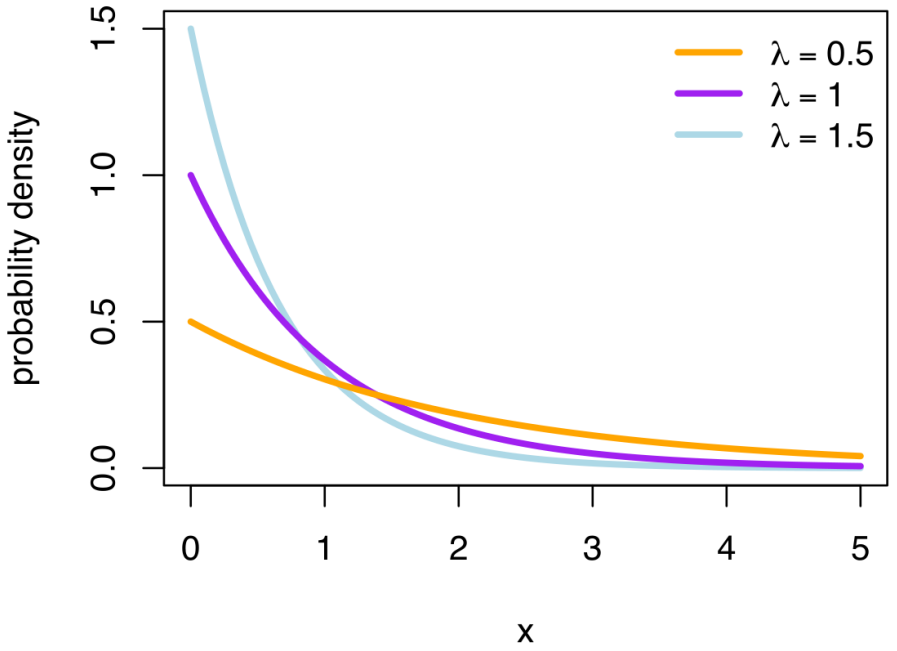
Distributions - exponential

Let $\lambda > 0$ be a fixed constant. Define a function f:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$E(X) = \frac{1}{\lambda}$$

$$var(X) = \frac{1}{\lambda^2}$$



Distributions – Gamma

The *gamma function* is defined by

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\, dt\,, \qquad \alpha > 0.$$

A property

$$\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$$

if n is a positive integer, then $\Gamma(n) = (n-1)!$

Gamma distribution:

$$f(x) = \frac{\lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)} e^{-\lambda x}$$

when $x > 0$, with $f(x) = 0$ for $x \leq 0$.

Gamma(1, λ) = Exponential(λ).

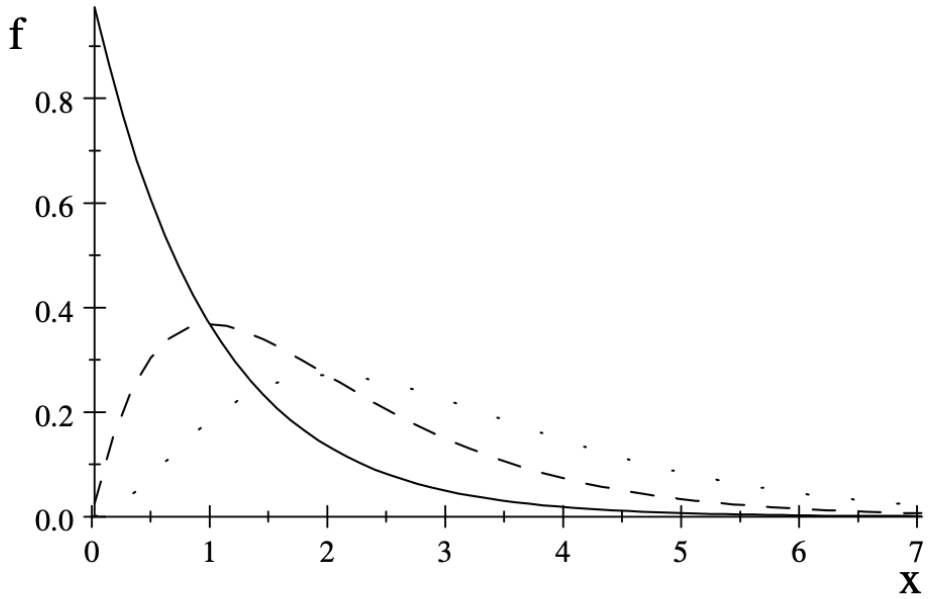


Figure 2.4.2: Graph of an Exponential(1) (solid line), a Gamma(2, 1) (dashed line), and a Gamma(3, 1) (dotted line) density.

Distributions – Gaussian (normal)

The standard normal N(0,1) distribution:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

The $N(\mu, \sigma^2)$ Distribution

$$f(x) = \frac{1}{\sigma}\,\phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/2\sigma^2}$$

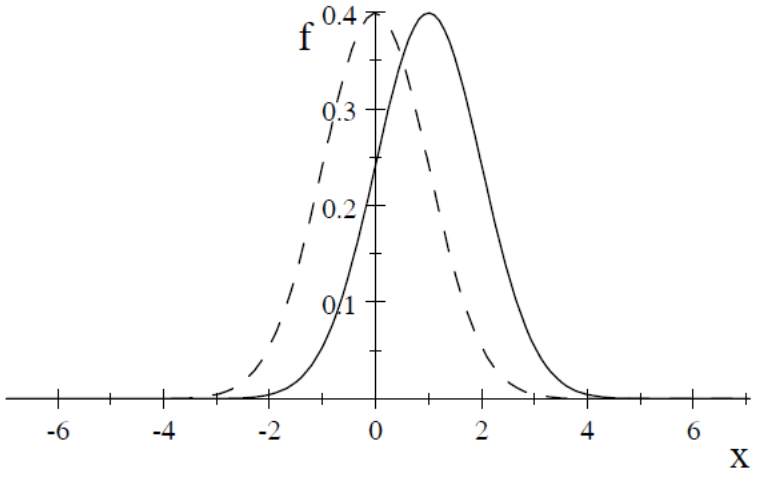


Figure 2.4.4: Graph of the $N(1, 1)$ density (solid line) and the $N(0, 1)$ density (dashed line).

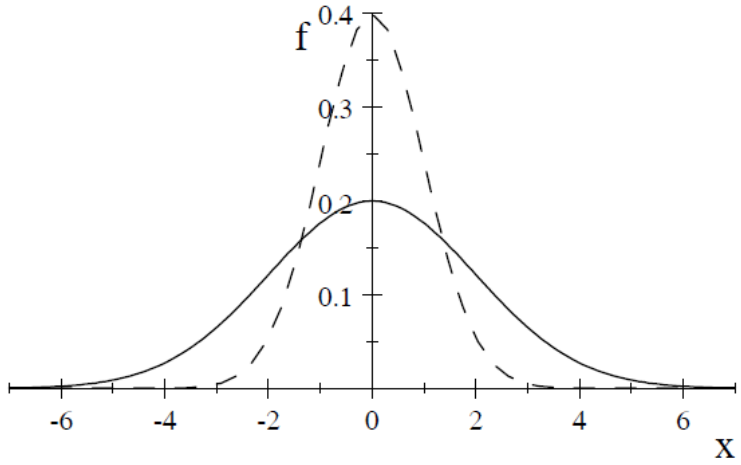


Figure 2.4.5: Graph of a $N(0, 4)$ density (solid line) and a $N(0, 1)$ density (dashed line).

Distributions – Chi-sq

Definition 4.6.1 The *chi-squared distribution* with n degrees of freedom (or chi-squared(n) , or $\chi^2(n)$) is the distribution of the random variable

$$Z = X_1^2 + X_2^2 + \cdots + X_n^2,$$

where $X_1, \ldots, X_n$ are i.i.d., each with the standard normal distribution $N(0, 1)$.

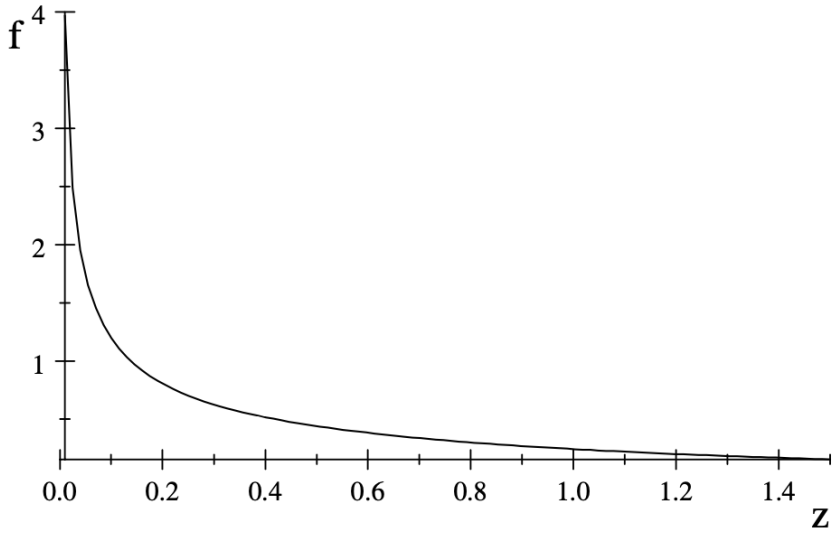


Figure 4.6.1: Plot of the $\chi^2(1)$ density.

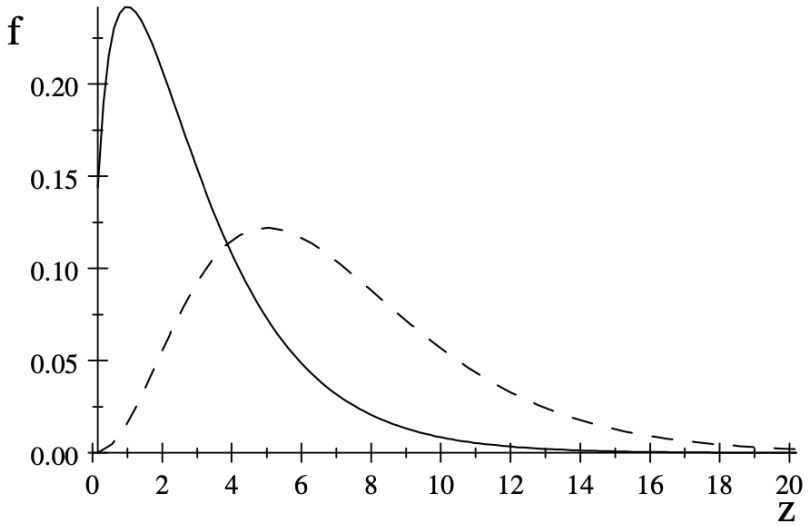


Figure 4.6.2: Plot of the $\chi^2(3)$ (solid line) and the $\chi^2(7)$ (dashed line) density functions.

$E(X) = n$

$var(X) = 2n$

Distributions – Chi-sq

Theorem 4.6.6 Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. Put

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n) \text{ and } S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Then $(n-1)\,S^2/\sigma^2 \sim \chi^2(n-1)$, and furthermore, S^2 and $\bar{X}$ are independent.

Distributions – t

Definition 4.6.2 The *t distribution* with n degrees of freedom (or Student(n), or $t(n)$), is the distribution of the random variable

$$Z = \frac{X}{\sqrt{(X_1^2 + X_2^2 + \cdots + X_n^2)/n}},$$

where $X, X_1, \ldots, X_n$ are i.i.d., each with the standard normal distribution $N(0, 1)$. (Equivalently, $Z = X/\sqrt{Y/n}$, where $Y \sim \chi^2(n)$.)

As $n \to \infty$, t(n) converges to the distribution of *Z (standard normal)*

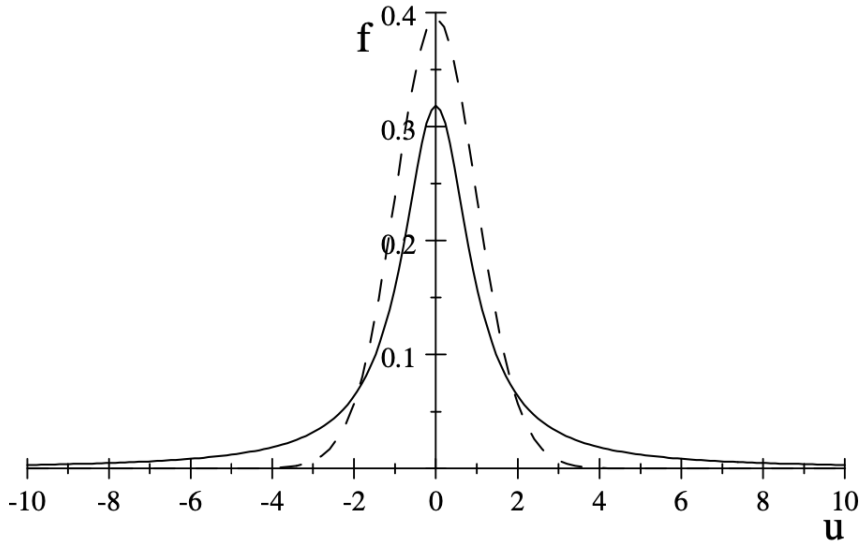


Figure 4.6.3: Plot of the $t(1)$ (solid line) and the $t(30)$ (dashed line) density functions.

Distributions – F

Definition 4.6.3 The *F distribution* with m and n degrees of freedom (or $F(m, n)$) is the distribution of the random variable

$$Z = \frac{(X_1^2 + X_2^2 + \cdots + X_m^2)\,/\,m}{(Y_1^2 + Y_2^2 + \cdots + Y_n^2)\,/\,n},$$

where $X_1, \ldots, X_m, Y_1, \ldots, Y_n$ are i.i.d., each with the standard normal distribution. (Equivalently, $Z = (X/m)/(Y/n)$, where $X \sim \chi^2(m)$ and $Y \sim \chi^2(n)$.)

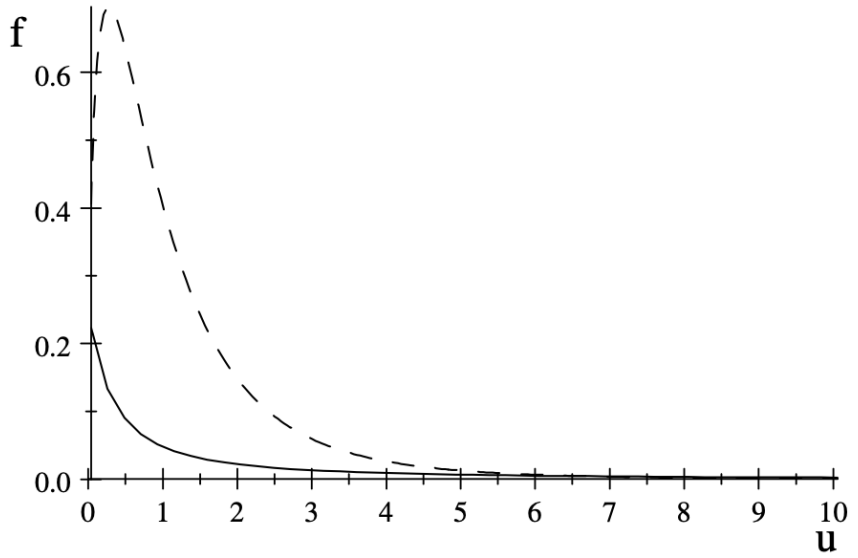


Figure 4.6.4: Plot of the $F(2, 1)$ (solid line) and the $F(3, 10)$ (dashed line) density functions.